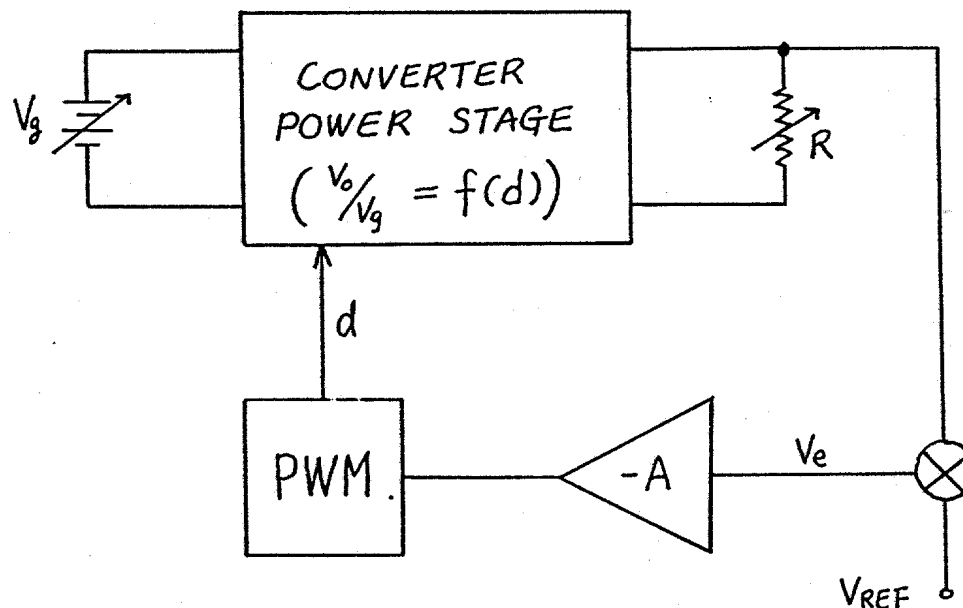


Chapter 3

Voltage-Mode Control for PWM Converters

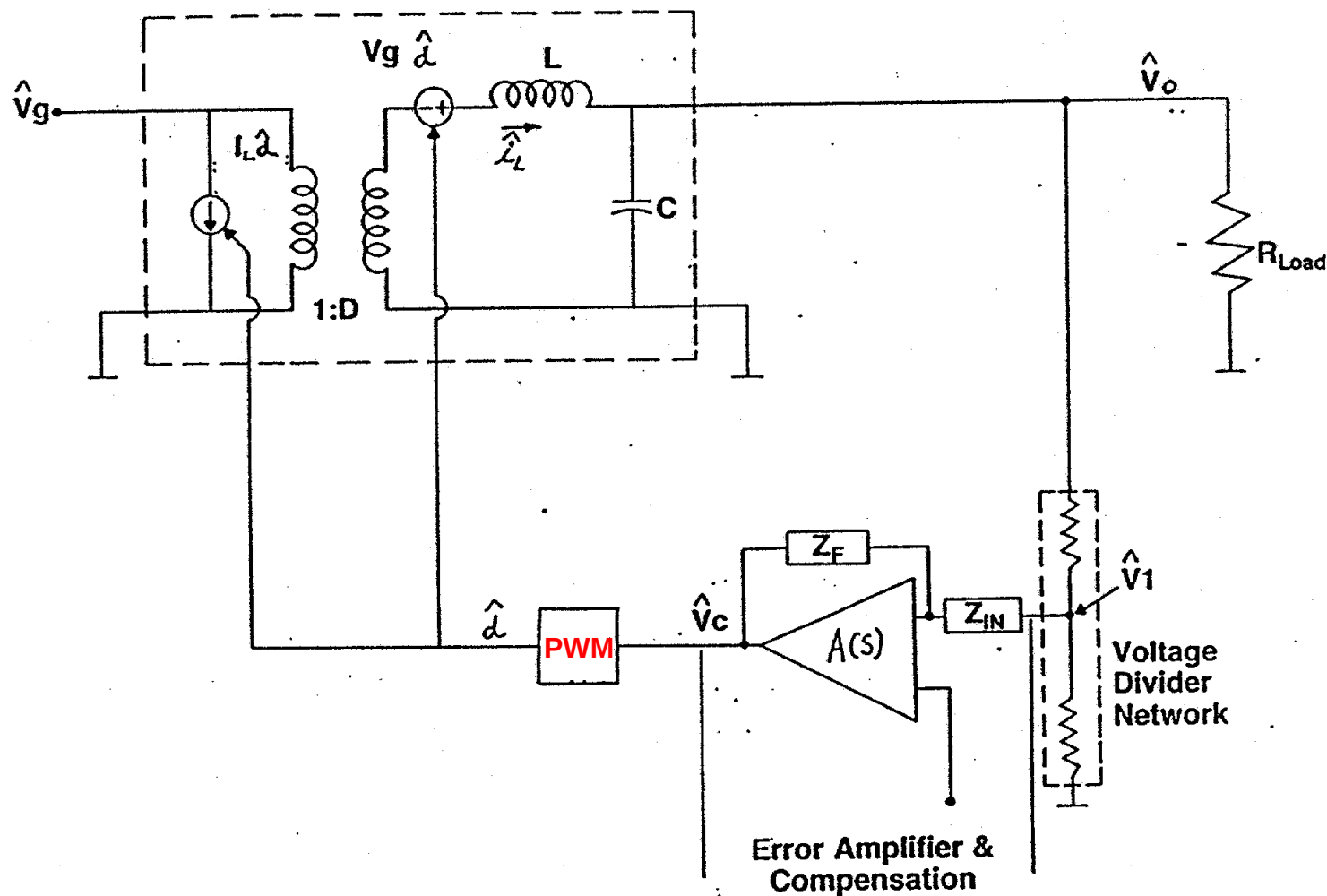
Single-Loop Controlled Switching Regulator



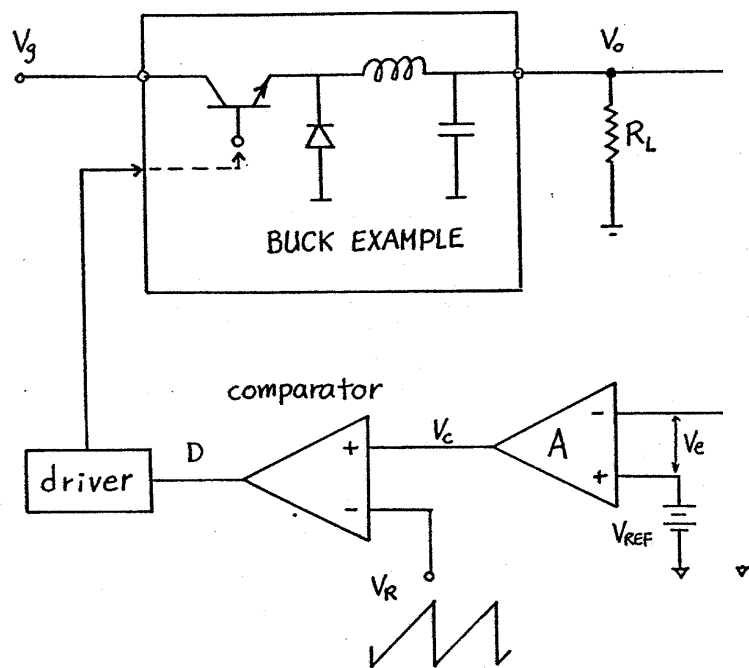
Feedback Control to Achieve:

- **Accuracy** High Gain, means T infinite large, $v_o/v_g \rightarrow 1$, better tracking
- **Speed** High bandwidth, if it is low, then no response on this frequency range
- **Stability**

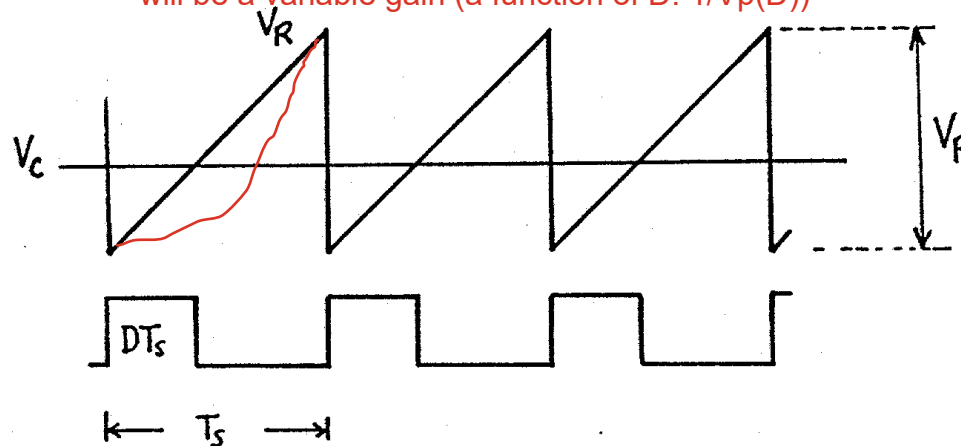
Small-Signal Analysis of Switching Regulators



DC PWM Gain

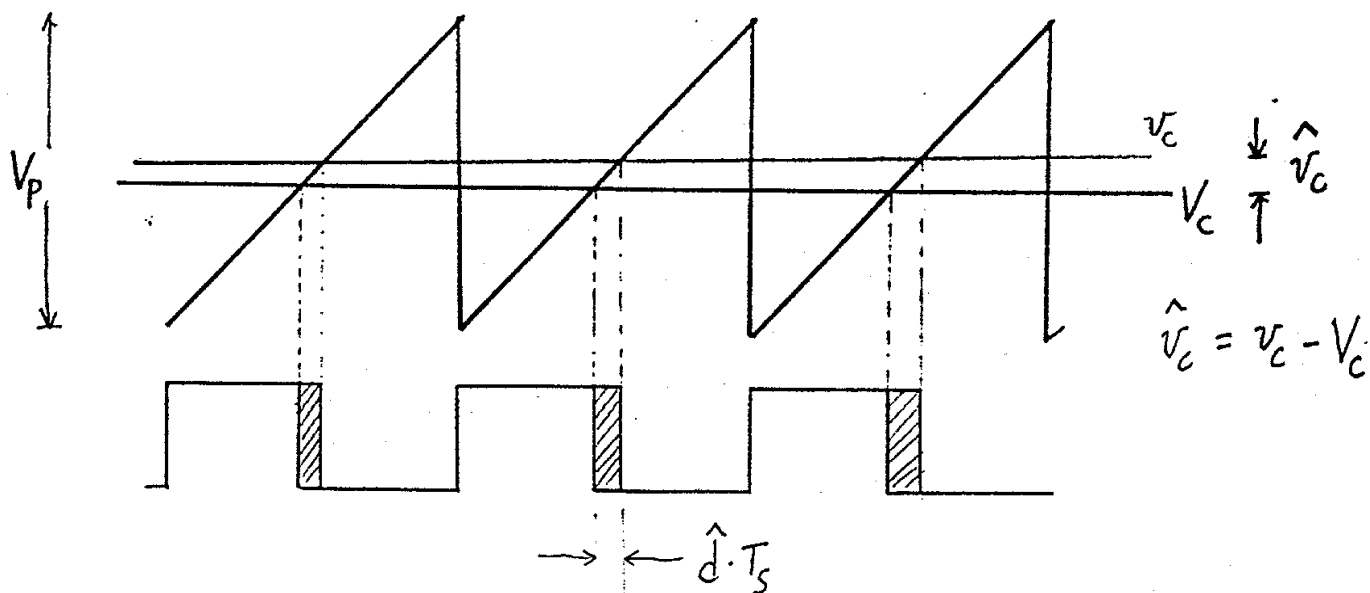


if the sawtooth modular change to non-constant slope sawtooth then the modular gain will not be a fixed gain ($1/V_p$), instead, it will be a variable gain (a function of D : $1/V_p(D)$)



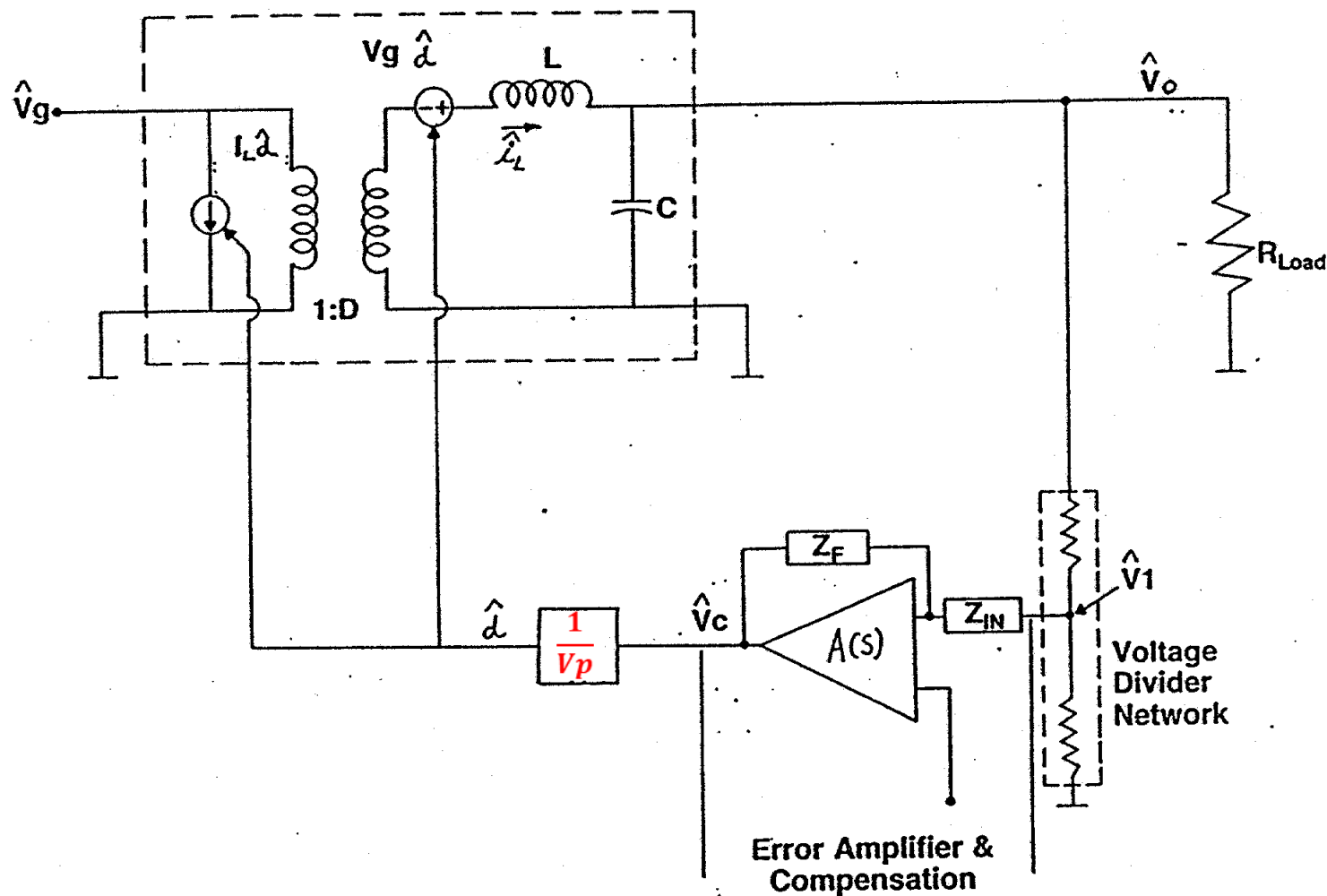
$$\text{DC PWM Gain : } \frac{D}{V_c} = \frac{1}{V_p}$$

Small-Signal PWM gain

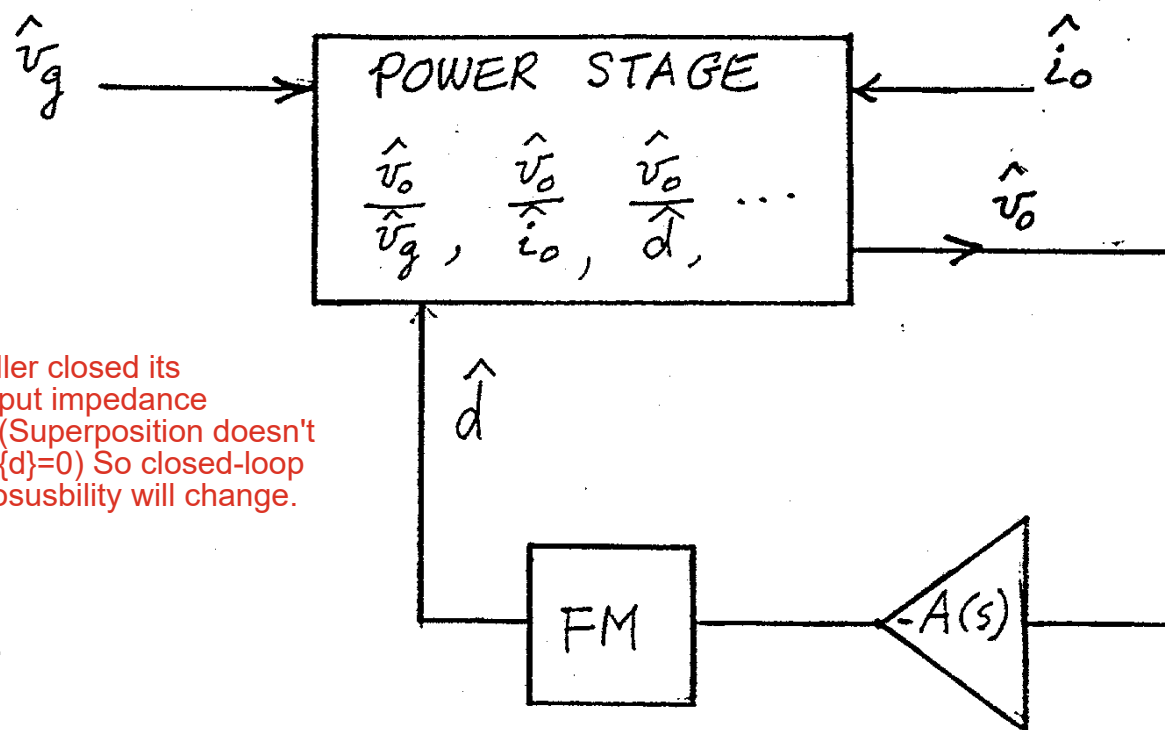


$$FM = \frac{\hat{d}}{\hat{v}_c} = \frac{1}{V_P}$$

Small-Signal Analysis of Switching Regulators



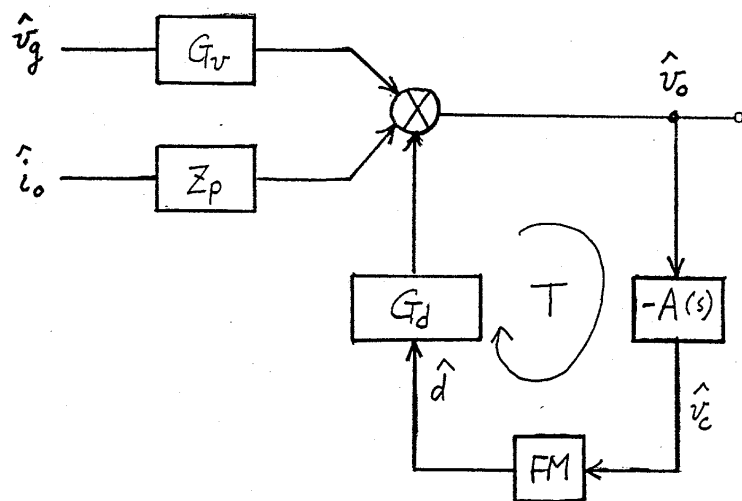
Small-Signal Analysis of Switching Regulators



when controller closed its loop, the output impedance will change. (Superposition doesn't work for $\hat{d}=0$) So closed-loop Z_o and audiosusibility will change.

Small-Signal Block Diagram

where:



Based on this diagram we can derive closed-loop output impedance and audiosusceptibility

$$G_v = \left. \frac{\hat{v}_o}{\hat{v}_g} \right|_{\hat{d}=0}$$

open-loop audio-susceptibility

$$Z_p = \left. \frac{\hat{v}_o}{\hat{i}_o} \right|_{\hat{d}=0}$$

open-loop output impedance

$$G_d = \left. \frac{\hat{v}_o}{\hat{d}} \right|_{\hat{v}_g=\hat{i}_o=0}$$

control-to-output transfer function

$$A(s) = \frac{\hat{v}_c}{\hat{v}_o}$$

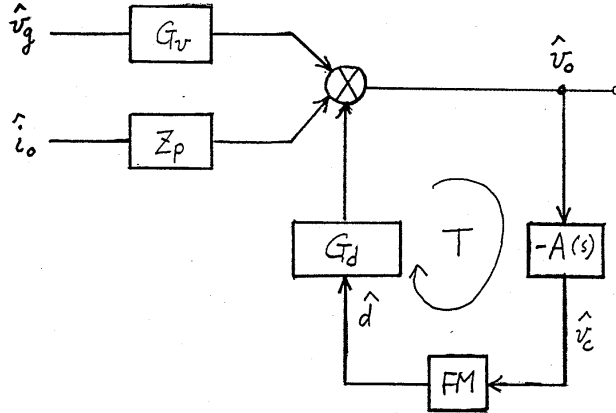
compensator gain

$$FM = \frac{\hat{d}}{\hat{v}_c}$$

PWM gain

Closed-Loop Transfer Functions

- Closed-loop audio susceptibility, $\frac{\hat{v}_o}{\hat{v}_g}$



$$\hat{v}_o = G_v \cdot \hat{v}_g + G_d \cdot \hat{d} \quad (1)$$

$$\hat{d} = -A \cdot FM \cdot \hat{v}_o \quad (2)$$

Eliminated $\hat{d}$ from (1) and (2)

push bandwidth and DC gain as high as possible,

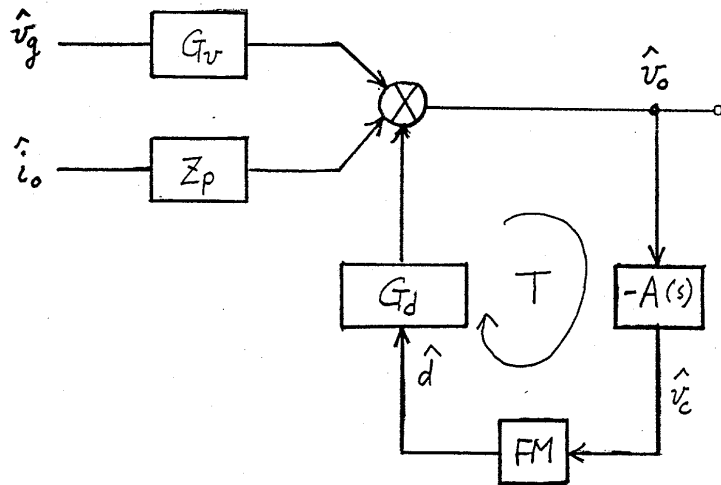
$$\Rightarrow \frac{\hat{v}_o}{\hat{v}_g} = \frac{G_v}{1 + G_d \cdot FM \cdot A} \stackrel{\Delta}{=} \boxed{\frac{G_v}{1 + T}}$$

when T is big enough, 1 is ignored, then $G_v/G_d = 1/V_g$, the system high freq gain will be determined pure by $FM \cdot A$

where $T = G_d \cdot FM \cdot A$: Loop gain

Closed-Loop Transfer Functions

- Closed-loop output impedance, $\frac{\hat{\mathbf{v}}_o}{\hat{\mathbf{i}}_o}$



$$\hat{\mathbf{v}}_o = \mathbf{Z}_p \cdot \hat{\mathbf{i}}_o + \mathbf{G}_d \cdot \hat{\mathbf{d}} \quad (3)$$

From (3) and (2)

$$\boxed{\frac{\hat{\mathbf{v}}_o}{\hat{\mathbf{i}}_o} = \frac{\mathbf{Z}_p}{\mathbf{1} + \mathbf{T}}}$$

Loop Gain Analysis

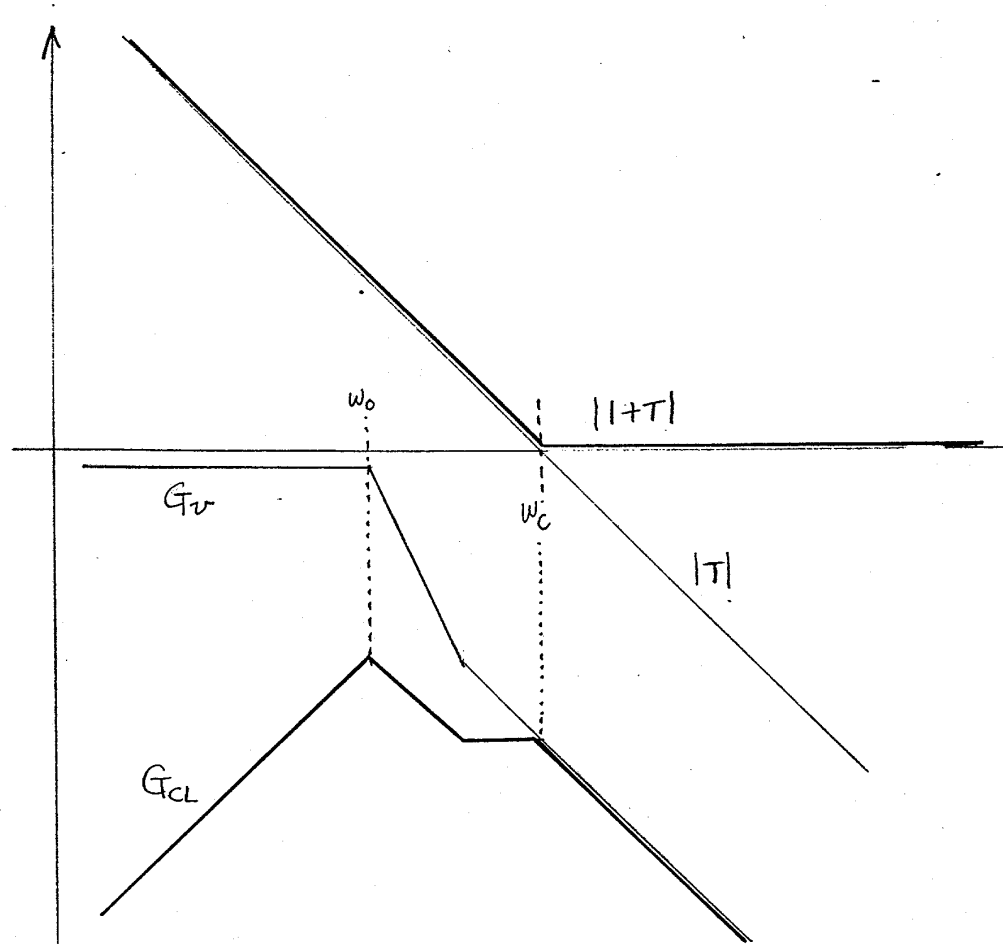
Provides:

- **System performance analysis**
- **Stability analysis**
 - **absolute stability**
 - **degree of stability**
- **Design insight**
- **Measurement verification**

Function of the Loop Gain, T

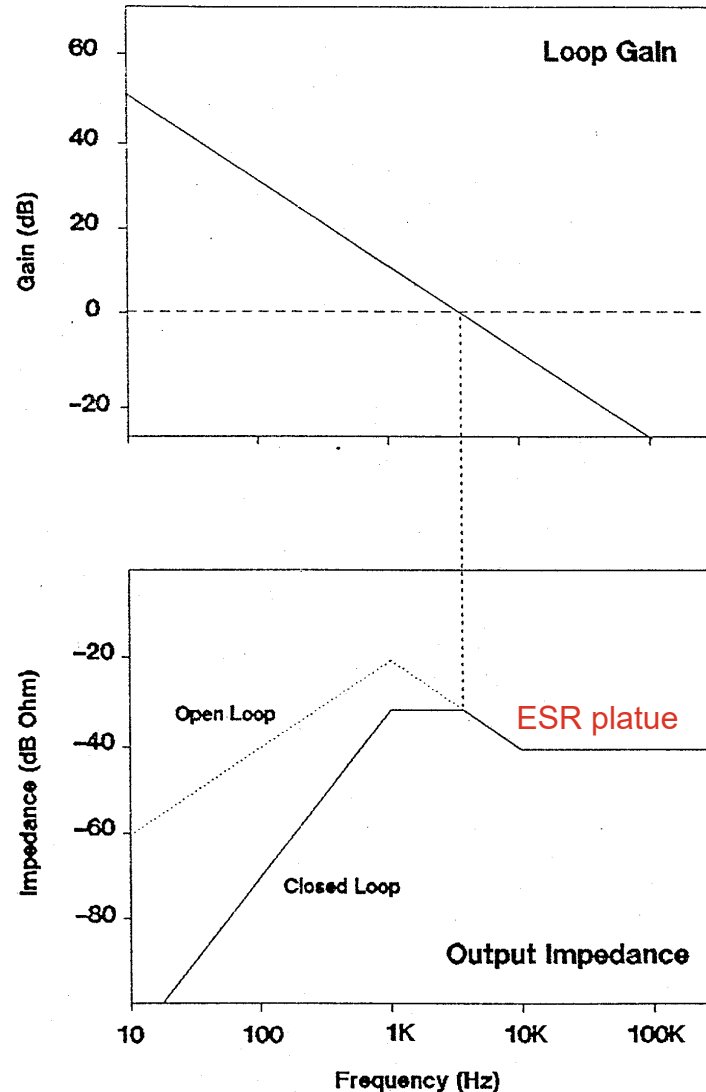
Example: Closed-loop audio-susceptibility

$$\left. \frac{\hat{V}_o}{\hat{V}_g} \right|_{CL} = \frac{G_v}{1+T} \stackrel{\Delta}{=} G_{CL} \text{ dB}$$

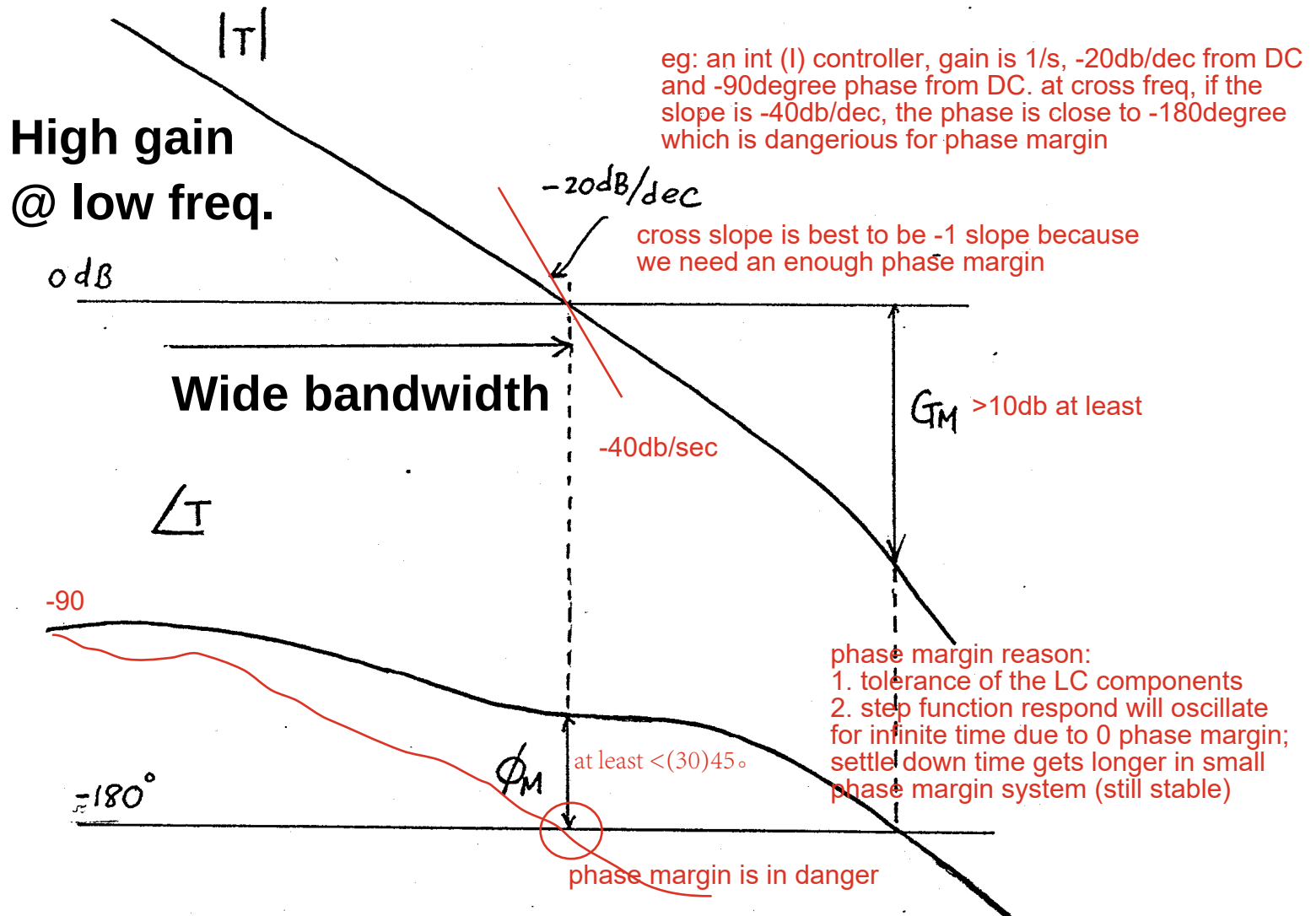


Function of the Loop Gain, T

Example: Closed-Loop Output Impedance



Desired Loop Gain Characteristics



Nyquist Criterion

$$Z = P - N$$

Z: # of RHP poles in T

P: # of RHP roots of $(1+T = 0)$

N: # of CW encirclement of $(-1, j0)$ point of the polar plot of T

For the stability, P must be zero

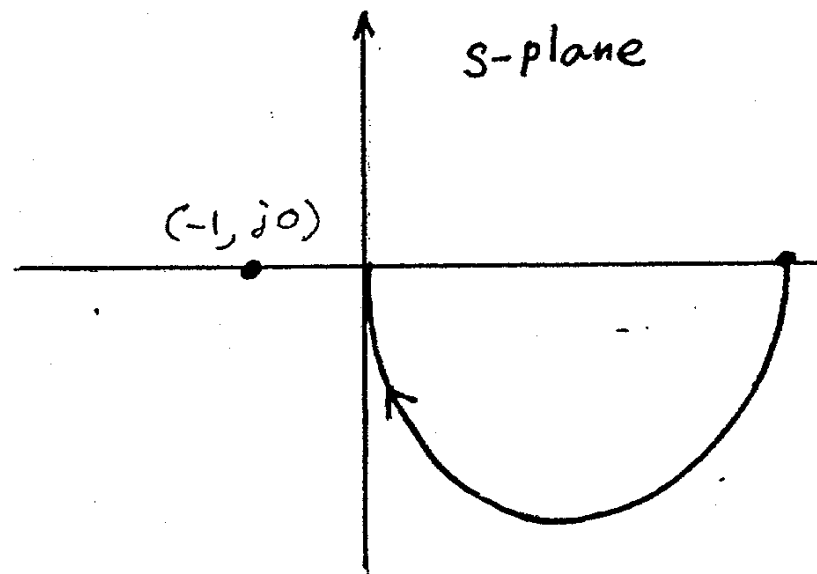
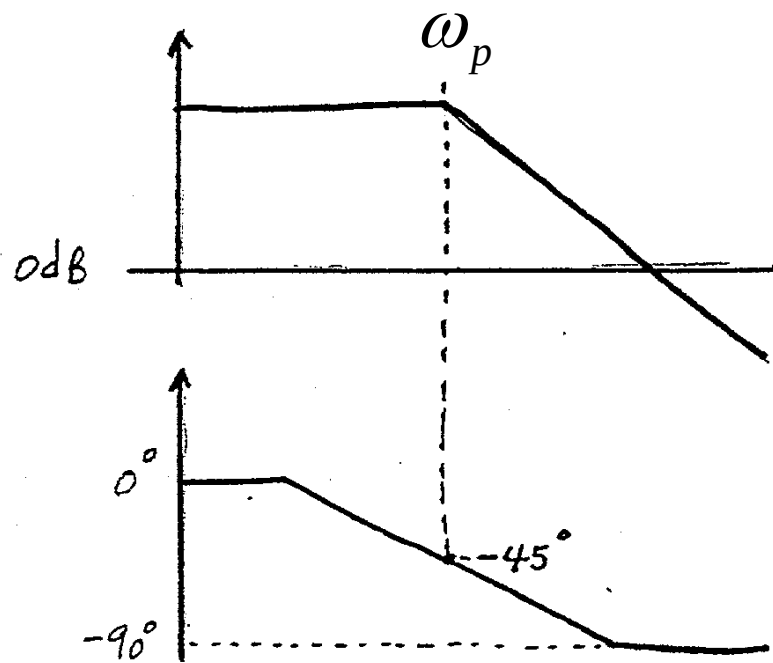
**** In practical systems, usually $Z = 0$**

=== Polar plot of T must **not encircle $(-1, j0)$ point for stability**

=== or in Bode plot, when the magnitude of T is 1 (0 dB), the phase of T should be greater than -180 degrees.

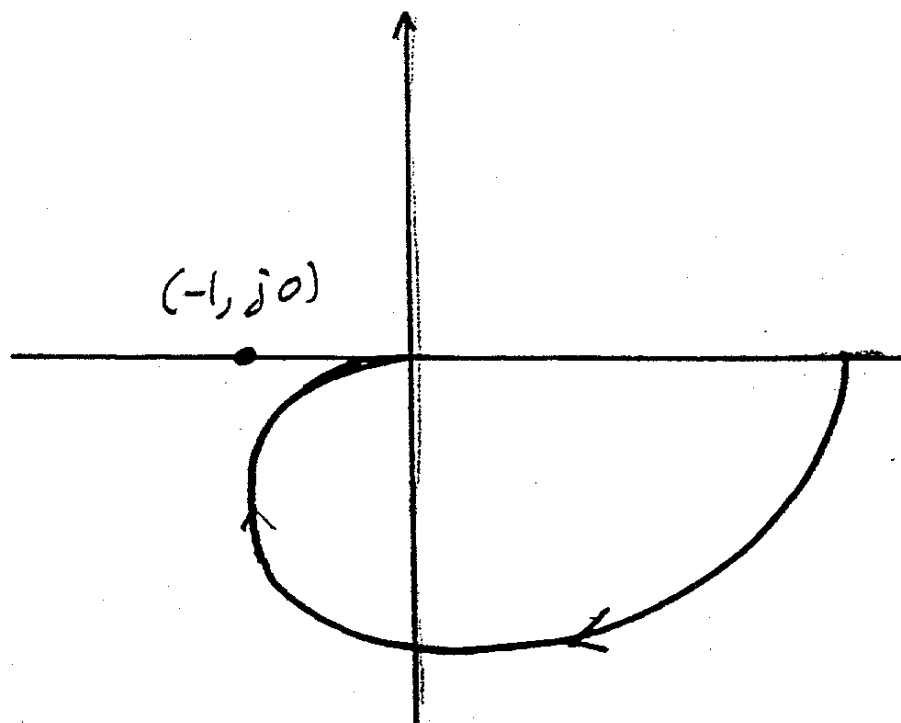
Examples

- 1st order system: $T = \frac{G_m}{1 + s/\omega_p}$

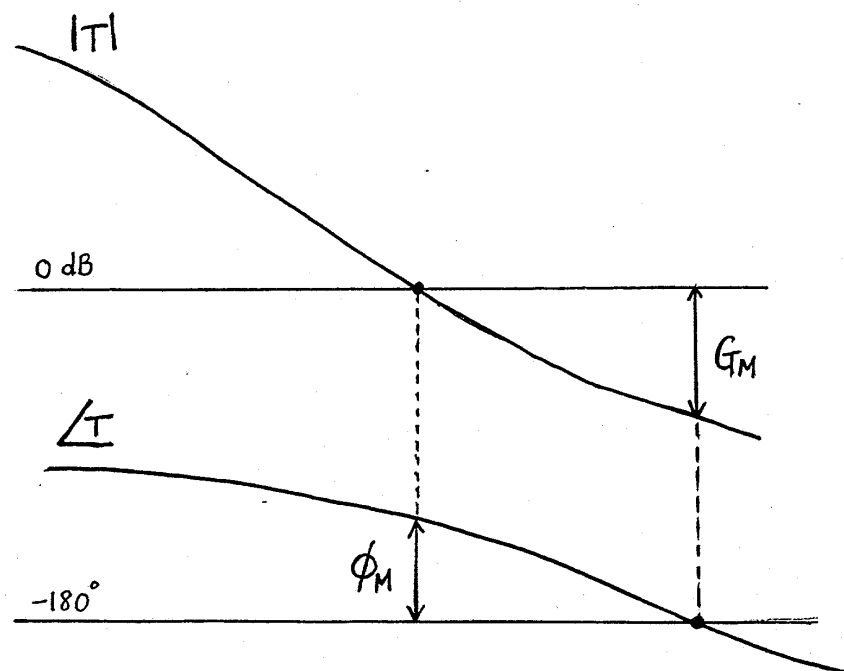
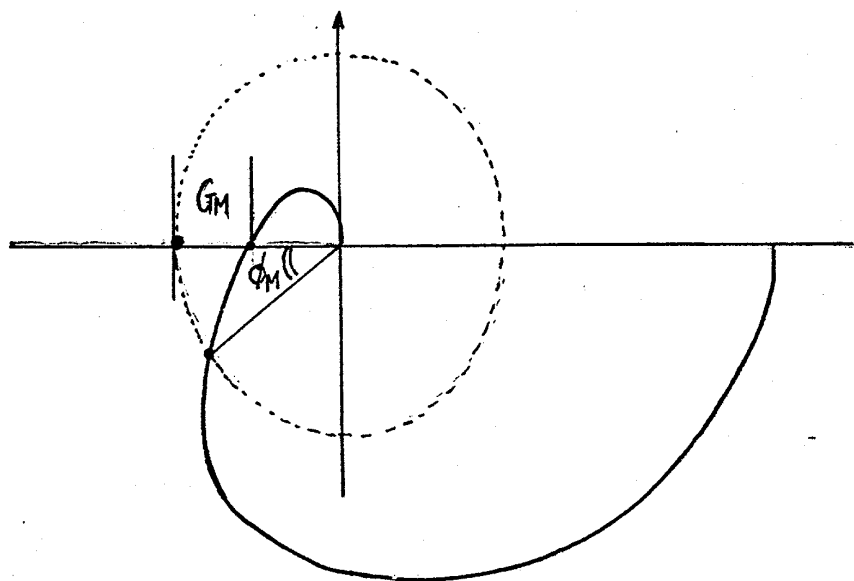


Examples

- 2nd order system: $T = \frac{G_m}{1 + s/Q\omega_0 + s^2/\omega_0^2}$



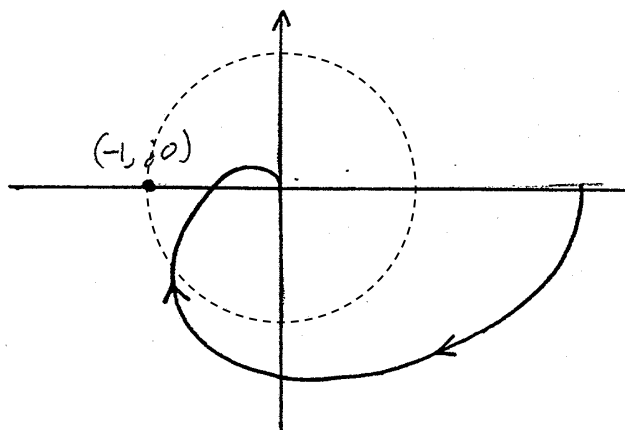
Stability Margins



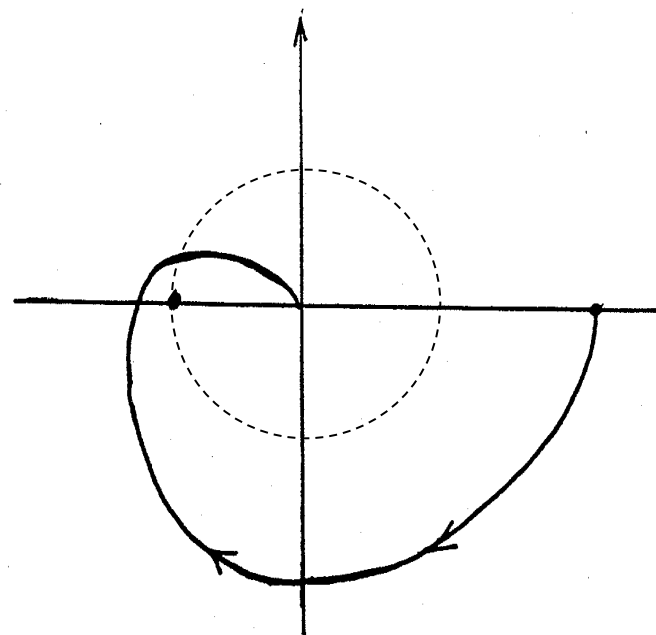
Examples

- High order system

Stable :



unstable :



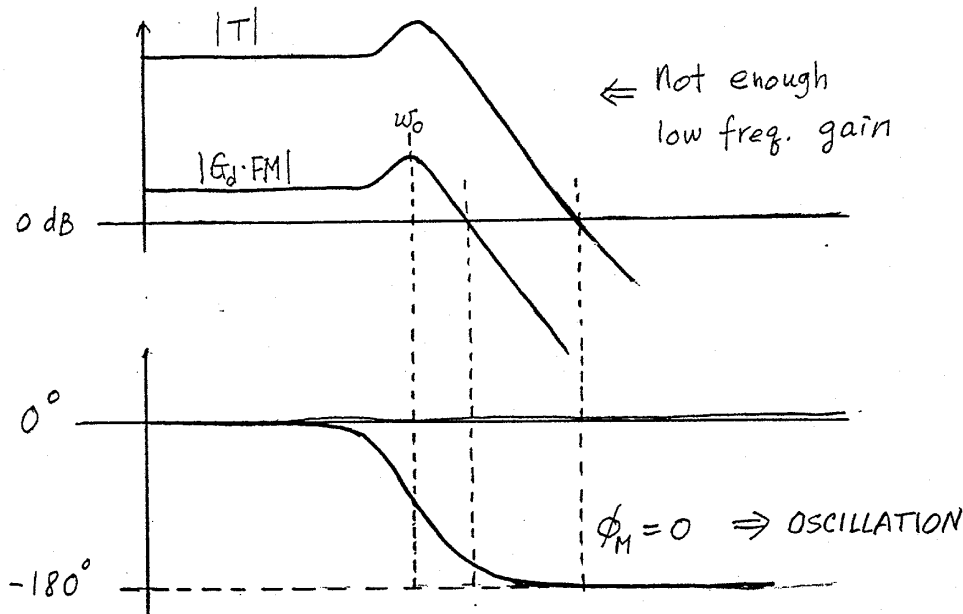
Compensator, $A(s)$ Design Considerations

— to Shape the Loop Gain, $T = G_d FM A(s)$

Consider an ideal buck converter

$$G_d = \frac{G_m}{1 + \frac{s}{Q\omega_0} + \left(\frac{s}{\omega_0}\right)^2}, \quad G_m = \frac{V_o}{D}, \quad \omega_0 = \frac{1}{\sqrt{LC}}, \quad Q = R\sqrt{\frac{C}{L}}$$

Suppose $A(s) = K$; constant gain



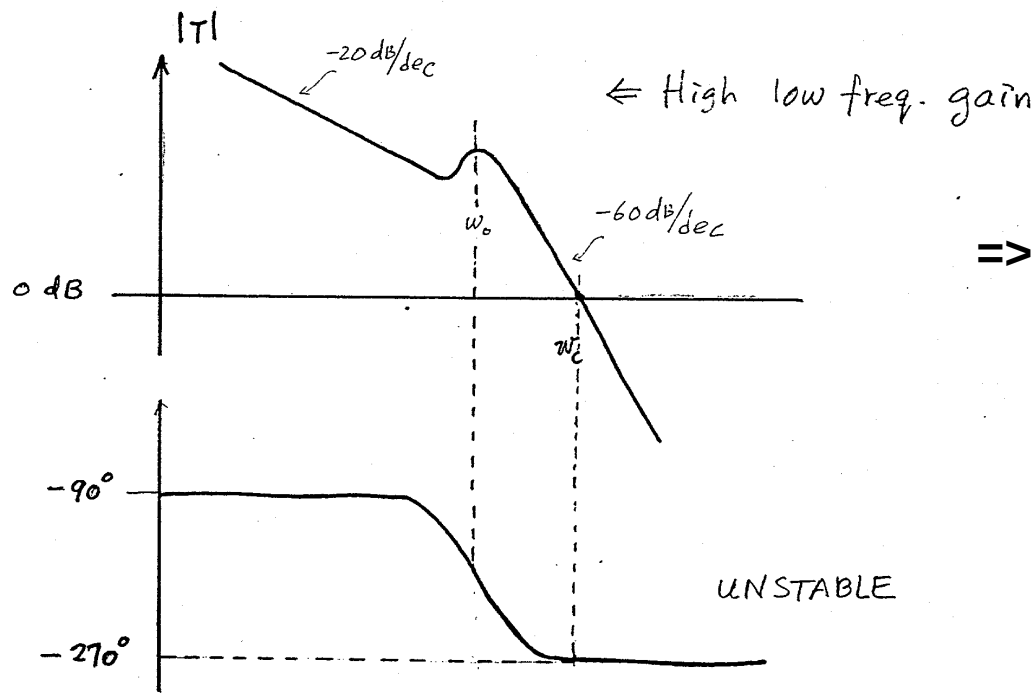
=> Need an integrator for a high gain at low frequency

Compensator, A(s) Design Considerations

— to Shape the Loop Gain, $T = G_d \text{ FM } A(s)$

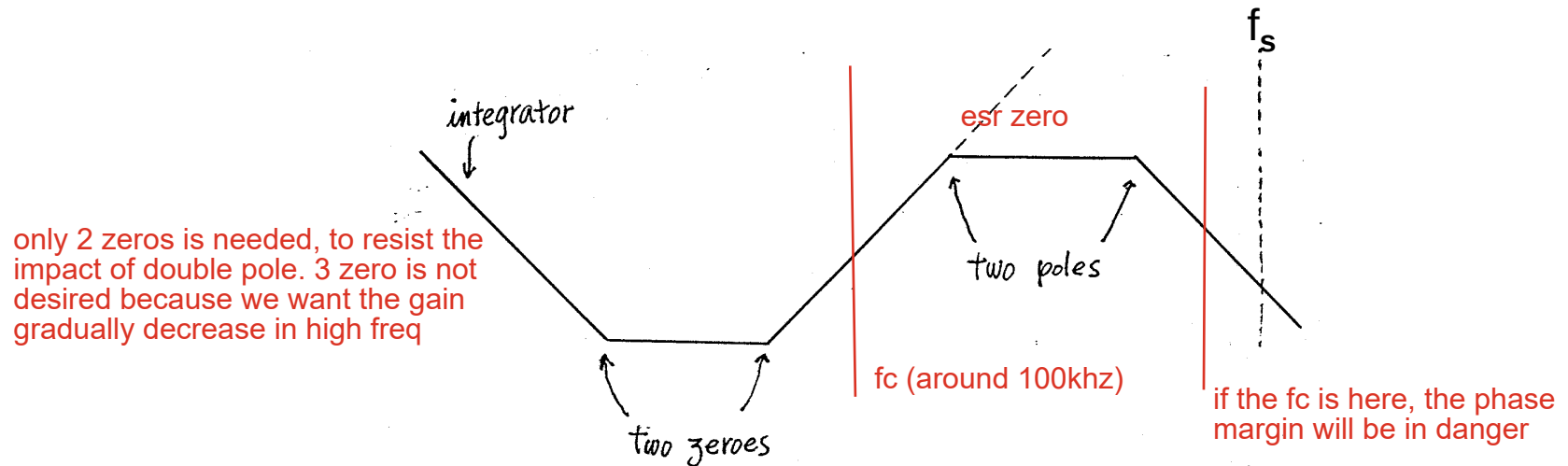
Suppose $A(s) = \frac{\omega_l}{s}$: integrator

$$T = \frac{\omega_m}{s} \cdot \frac{1}{1 + \frac{s}{Q\omega_0} + \left(\frac{s}{\omega_0}\right)^2}, \quad \omega_m = \omega_l \cdot G_m \cdot \text{FM}$$



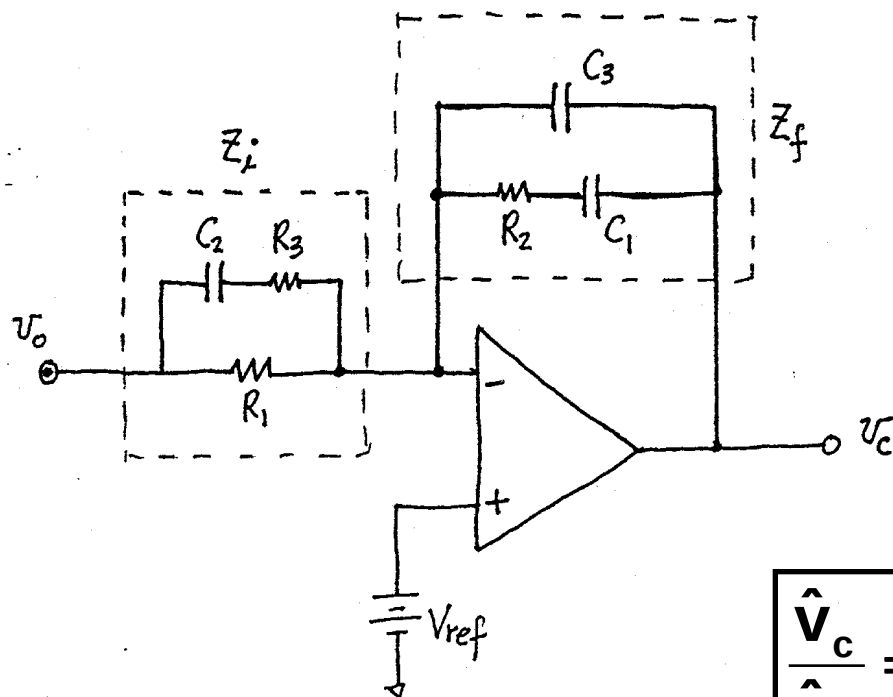
$\Rightarrow$ Excessive phase lag at the loop gain crossover, ω_c due to the integrator (-90°) and complex pole pair (-180°) need to be compensated by introducing two zeroes before ω_c

Compensator Characteristics



- An integrator for high dc gain
- Two zeroes below the loop gain crossover frequency, f_c to compensate the excessive phase lag due to the integrator and the power stage **complex pole pair**.
double pole
- Two high frequency pole
 - to attenuate high frequency noise
 - to ensure the magnitude of the loop gain keeps decreasing after the 0 dB crossover
 - phase lag due to the two poles at the loop gain crossover should be minimum

Compensator Circuit



$$\frac{\hat{V}_c}{\hat{V}_o} = -\frac{Z_f}{Z_i}$$

$$Z_i = R_1 \parallel \left(\frac{1}{sC_2} + R_3 \right)$$

$$Z_f = \frac{1}{sC_3} \parallel \left(\frac{1}{sC_1} + R_2 \right)$$

$$\frac{\hat{V}_c}{\hat{V}_o} = -\frac{\omega_l (1 + s/\omega_{z1})(1 + s/\omega_{z2})}{s (1 + s/\omega_{p1})(1 + s/\omega_{p2})}$$

where:

$$\omega_{z1} = \frac{1}{R_2 C_1}, \quad \omega_{z2} = \frac{1}{C_2 (R_1 + R_3)}$$

$$\omega_l = \frac{1}{R_1 (C_1 + C_3)}$$

$$\omega_{p1} = \frac{1}{R_3 C_2}, \quad \omega_{p2} = \frac{1}{R_2 \cdot \frac{C_1 C_3}{C_1 + C_3}}$$

Compensator Circuit

For $C_1 \gg C_3$, $R_1 \gg R_3$

$$\omega_l = \frac{1}{R_1 C_1}$$

$$\omega_{z1} = \frac{1}{R_2 C_1}, \quad \omega_{z2} = \frac{1}{R_1 C_2}$$

$$\omega_{p1} = \frac{1}{R_3 C_2}, \quad \omega_{p2} = \frac{1}{R_2 C_3}$$

Buck Regulator Design

Loop gain, $T = G_d \cdot FM \cdot A(s)$

$$\bullet \quad G_d = \frac{\Delta \hat{V}_o}{\hat{d}} = V_g \cdot \frac{1 + s/\omega_z}{1 + \frac{s}{Q\omega_0} + \left(\frac{s}{\omega_0}\right)^2}$$

$$\omega_z = \frac{1}{R_c C}$$

$$\omega_0 \approx \frac{1}{\sqrt{LC}}$$

$$Q = R \sqrt{\frac{C}{L}}$$

$$\bullet \quad G_v = \frac{\Delta \hat{V}_o}{\hat{V}_g} = D \frac{1 + s/\omega_z}{1 + \frac{s}{Q\omega_0} + \left(\frac{s}{\omega_0}\right)^2}$$

$$\bullet \quad Z_P = \frac{\Delta \hat{V}_o}{\hat{i}_o} = (R_l // R) \frac{(1 + s/\omega_z)(1 + s/\omega_{zL})}{1 + \frac{s}{Q\omega_0} + \left(\frac{s}{\omega_0}\right)^2}$$

$$\omega_{zL} = \frac{R_l}{L}$$

Loop Gain Design Procedure

$$T = \frac{\omega_m}{s} \cdot \frac{(1 + s/\omega_{z1})(1 + s/\omega_{z2})}{(1 + s/\omega_{p1})(1 + s/\omega_{p2})} \cdot \frac{(1 + s/\omega_z)}{\left(1 + \frac{s}{Q\omega_0} + \left(\frac{s}{\omega_0}\right)^2\right)}$$

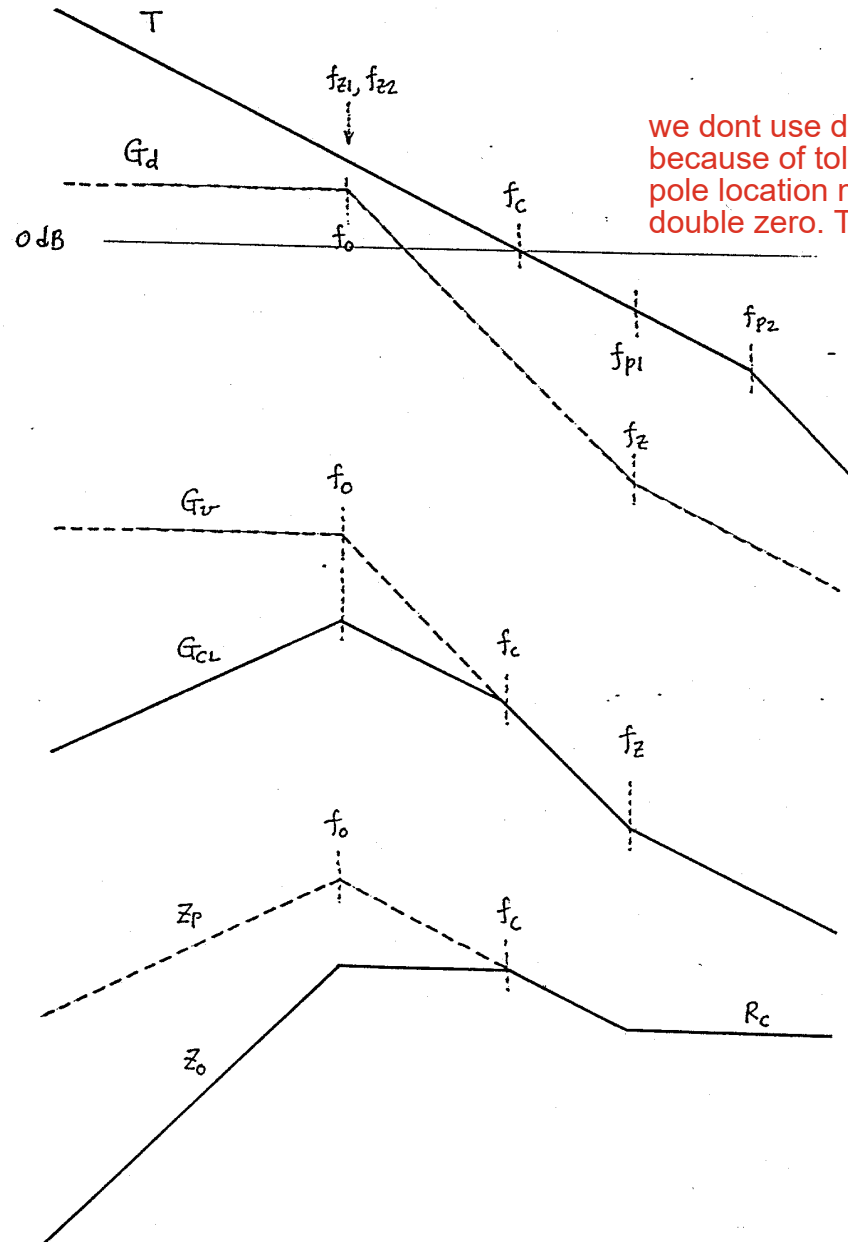
where $\omega_m = V_g \cdot FM \cdot \omega_l$

The objective is to design a compensator, ω_l , ω_{z1} , ω_{z2} , ω_{p1} and ω_{p2} to shape the loop gain, T for stability and optimum performance for given power stage parameters, ω_z , ω_0 , Q and V_g and the PWM gain, FM .

Design Steps

1. Set the loop gain crossover frequency, f_c for given switching frequency, f_s . i.e., $f_c = f_s / 6^{1/10 - 1/3}$
2. Cancel the ESR zero, f_z , by a compensator pole, f_{p1} . ($f_{p1} = f_z$)
3. Place a high frequency compensator pole, f_{p2} to get the maximum attenuation of the switching ripple and high frequency noise with the minimum phase lag at f_c . i.e., $f_{p2} \approx 10 f_c$
4. Place the two compensator zeroes, f_{z1} and f_{z2} below f_c .
5. The integrator gain, ω_I is determined after step 4.
6. Select the compensator parameters. (R's and C's)

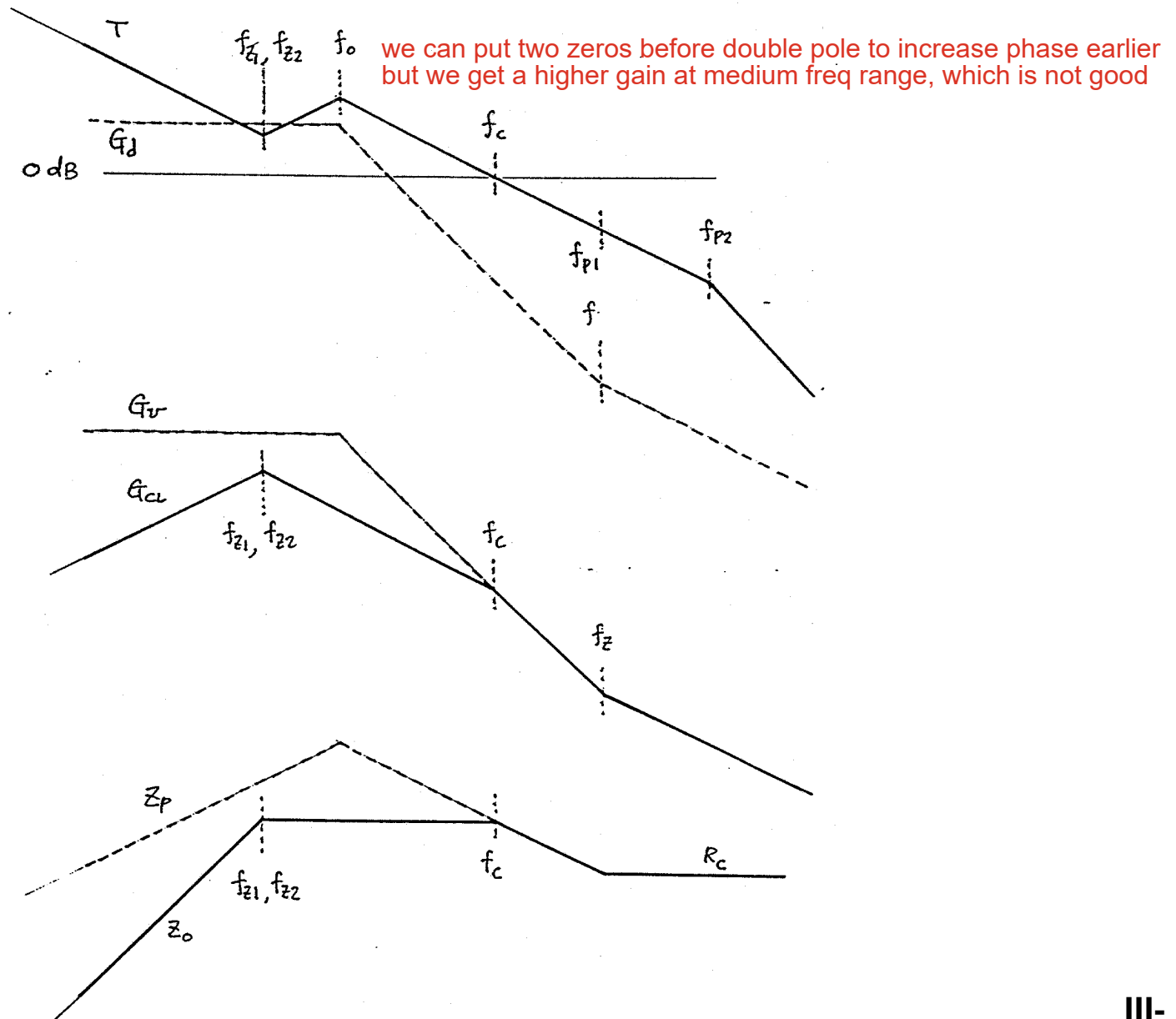
Design of f_{z1} and f_{z2} - Case 1



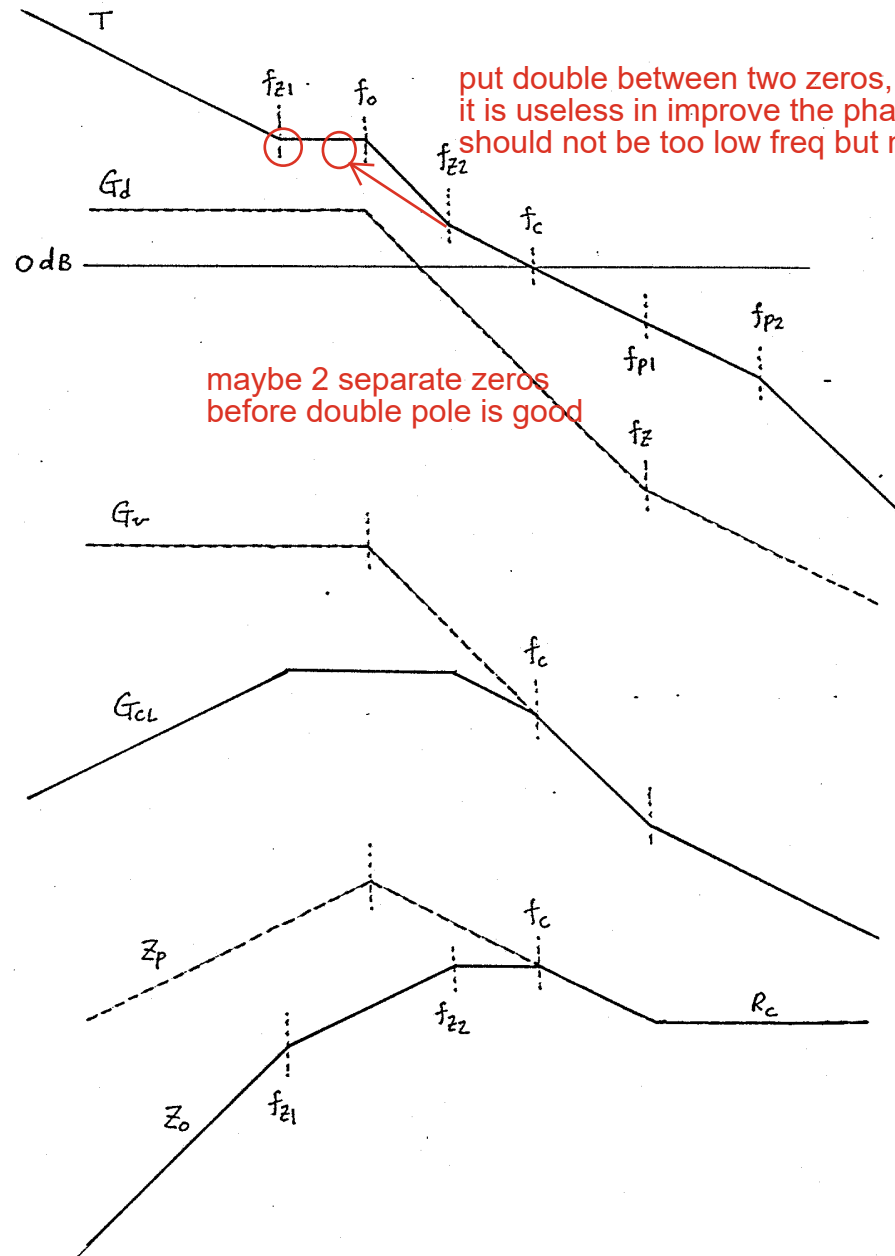
we dont use double zero to cancel double pole, because of toleranc of LC hardware, the double pole location may shift away from designed double zero. Then the oscillation may exist

double pole phase decrease quickly (-180) if no damping exist, double pole may not be able to level up the phase on f_c location

Design of f_{z1} and f_{z2} - Case 2



Design of f_{z1} and f_{z2} - Case 3



Settling Time for the Step Input / Load Transient

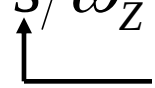
Closed-loop transfer function

$$G_{CL} = \frac{G_o}{1+T} \approx \frac{G_o}{T} \text{ at low frequency } (|T| \gg 1)$$

$$G_o = \frac{F_1(s)}{G(s)}$$

$$T = \frac{(1+s/\omega_z)}{G(s)} \frac{\omega_m}{s} \frac{(1+s/\omega_{z1})(1+s/\omega_{z2})}{(1+s/\omega_{p1})(1+s/\omega_{p2})}$$

$$\therefore G_{CL} \approx \frac{G_o}{T} = \frac{s/\omega_m \cdot F_1(s) \cdot (1+s/\omega_{p1})(1+s/\omega_{p2})}{(1+s/\omega_{z1})(1+s/\omega_{z2})(1+s/\omega_z)}$$


Come from ESR zero

The lower frequency compensator zero, f_{z1} determines the settling time,

Design Trade-Off for f_{z1} and f_{z2}

	freq.	closed-loop peak	Settling time	stability margin
f_{z1}	↑	↓	faster	↓
	↓	↑	slower	↑
f_{z2}	↑	↓		↓
	↓	↑		↑

Flyback Regulator Design

RHPZ: if exist in high freq, no problem for system
if exist in low freq, we can only reduce bandwidth

$$\bullet \quad G_d = \frac{\Delta \hat{v}_o}{\hat{d}} = \frac{N_s}{N_p} \cdot \frac{V_g}{D'^2} \cdot \frac{(1 - s/\omega_a)(1 + s/\omega_b)}{\left\| 1 + \frac{s}{Q\omega_0} + \left(\frac{s}{\omega_0} \right)^2 \right\|}$$

$$\omega_0 \approx \frac{D'}{\sqrt{LC}}, \quad Q \approx D'R \cdot \sqrt{\frac{C}{L}}$$

$$\omega_a \approx \frac{D'^2 R}{DL}, \quad \omega_b = \frac{1}{R_C C}$$

$$\bullet \quad G_v = \frac{\Delta \hat{v}_o}{\hat{v}_g} = \frac{D}{D'} \cdot \frac{N_s}{N_p} \cdot \frac{1 + s/\omega_b}{\left\| \right\|}$$

from CCM → DCM
then RHPZ will go to high freq

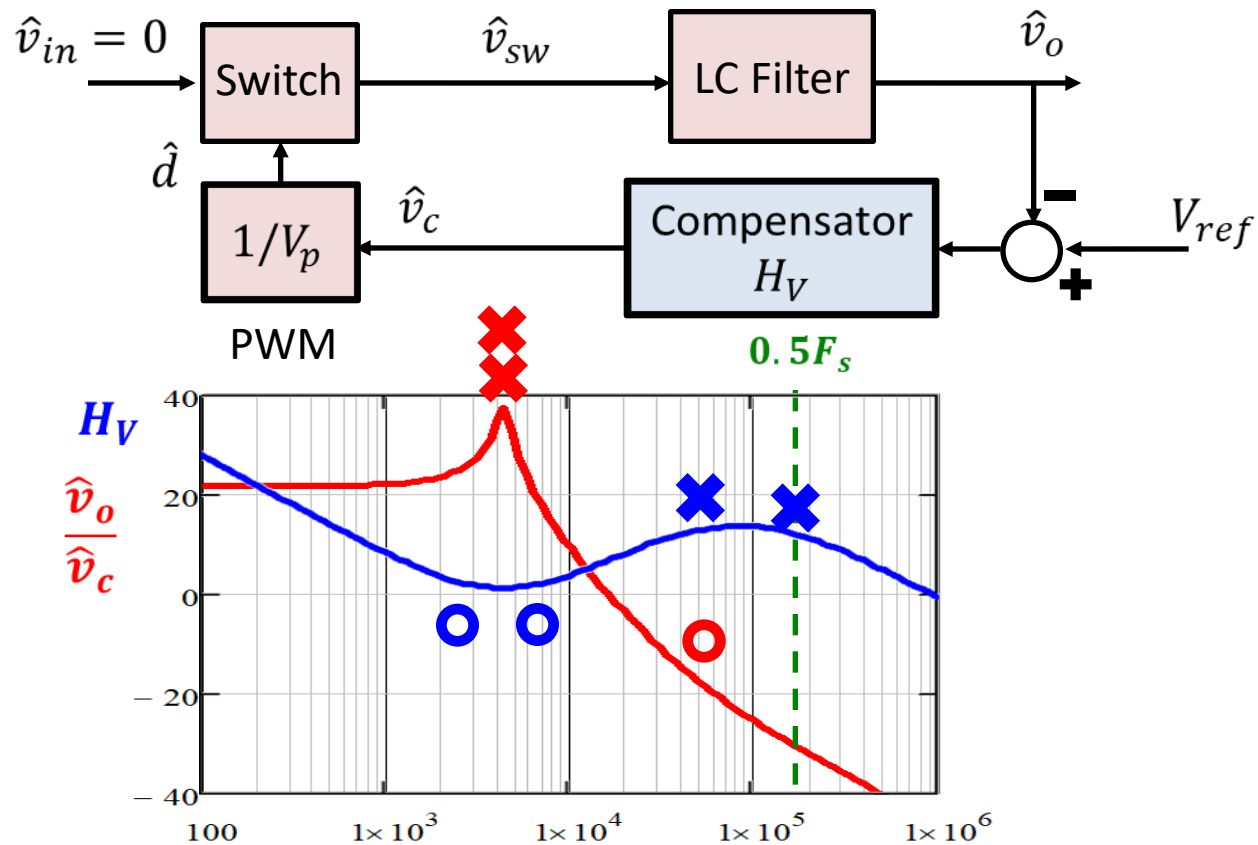
$$\bullet \quad Z_P = \frac{\Delta \hat{v}_o}{\hat{i}_o} = \left(\left(\frac{R_l}{D'^2} \right) // R \right) \cdot \frac{(1 + s/\omega_b)(1 + s/\omega_c)}{\left\| \right\|} \quad \omega_c = \frac{R_l}{L}$$

Loop Gain Design

- Due to the RHP zero, f_a the loop gain crossover frequency, f_c must be designed well below the RHP zero.
i.e., $f_c = f_a / 10$
- The two poles in the compensator should cancel both ESR zero, f_b and RHP zero, f_a
- Design of the two zeroes are the same as in the buck regulator case.

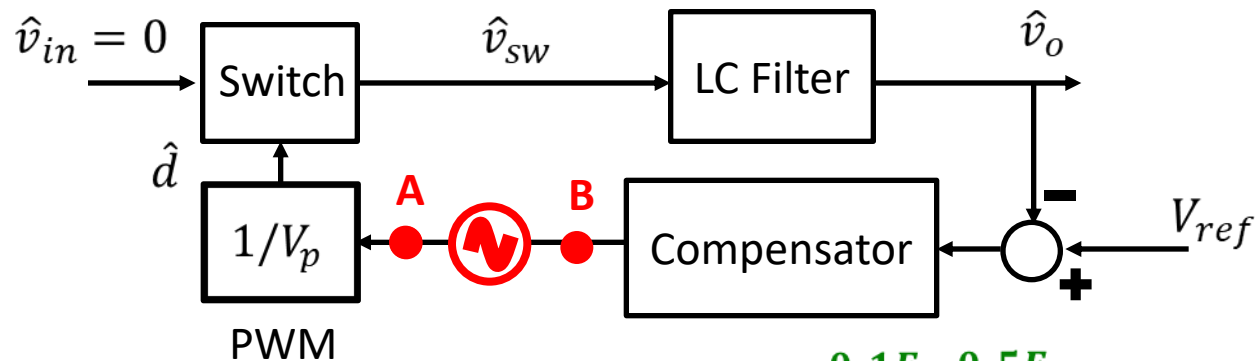
Bandwidth Impact on the Model Accuracy for Voltage Model Control

(Example: Buck Converter)



Prediction by the Average Modeling

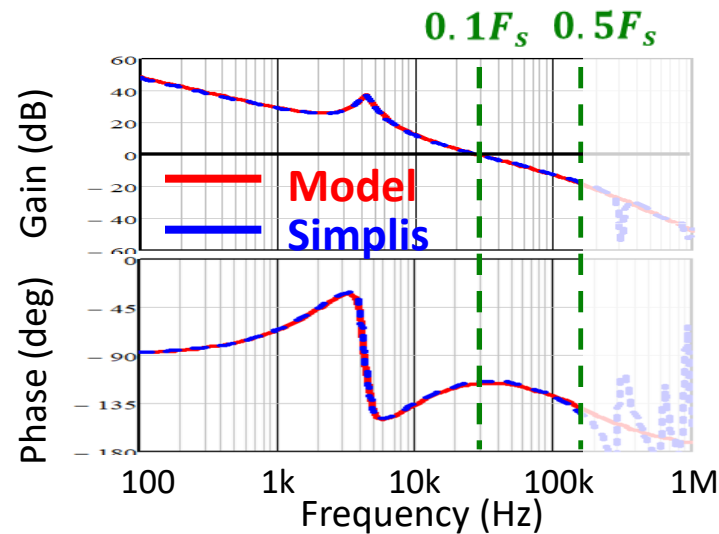
(low Bandwidth Example)



Closed-loop gain

$$\frac{v_B(f_m)}{v_A(f_m)}$$

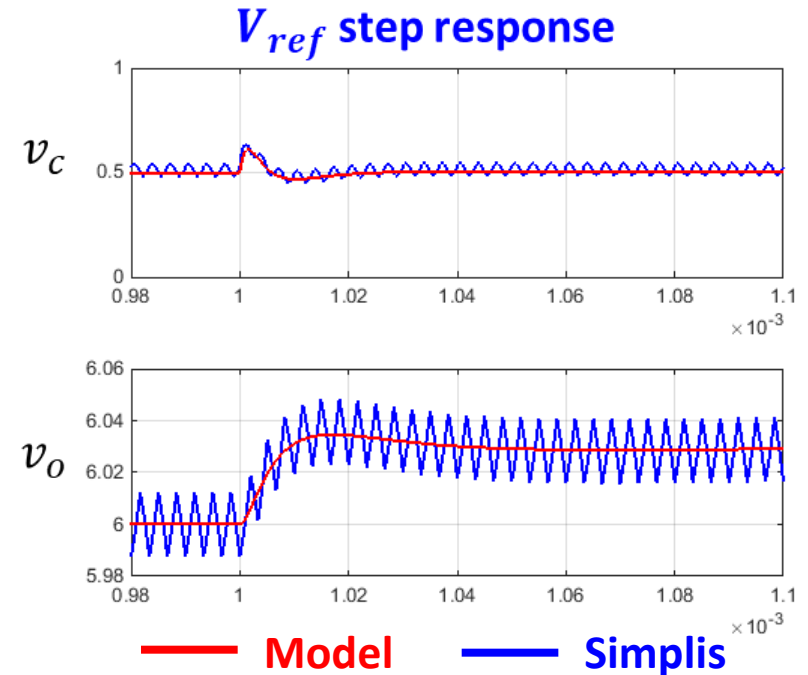
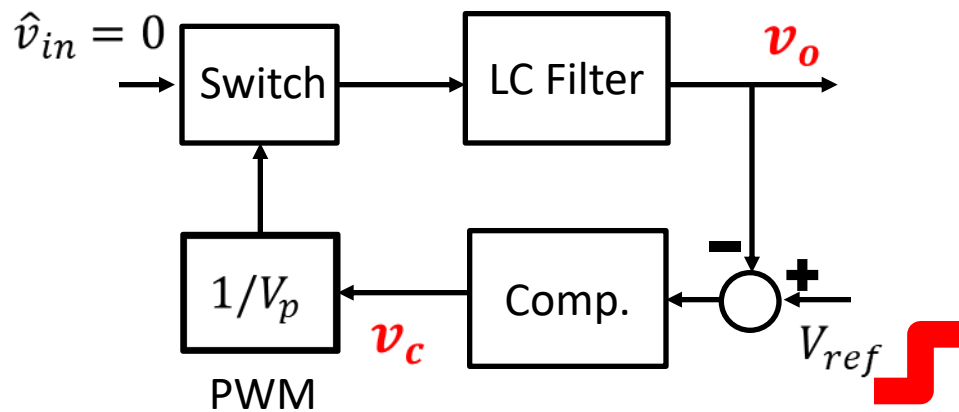
(Bandwidth = $0.1F_s$)



after half of f_s , model is above the bandwidth so this is not a issue

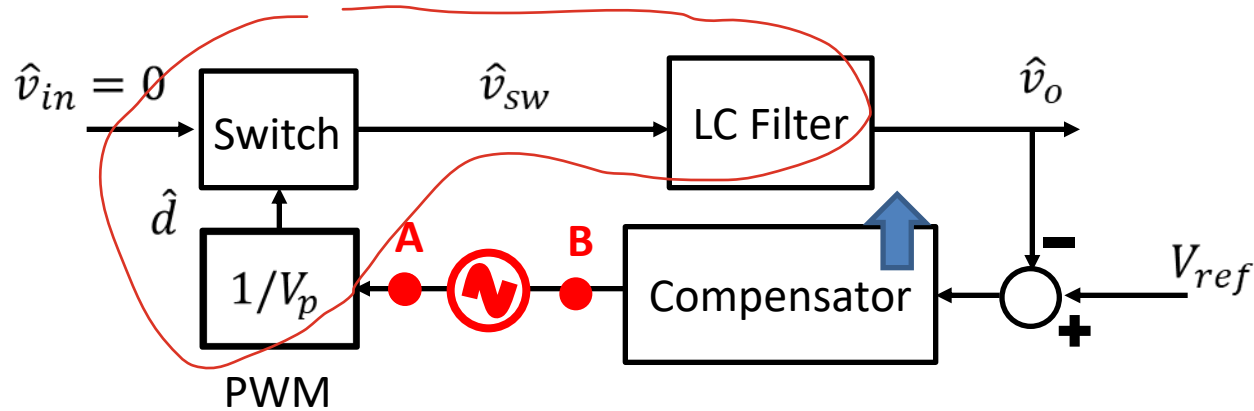
Average model is quite accurate with low bandwidth design

Prediction by the Average Modeling



Due to sufficient filtering, the average model well predicts the low-frequency dynamic of V_c and V_o .

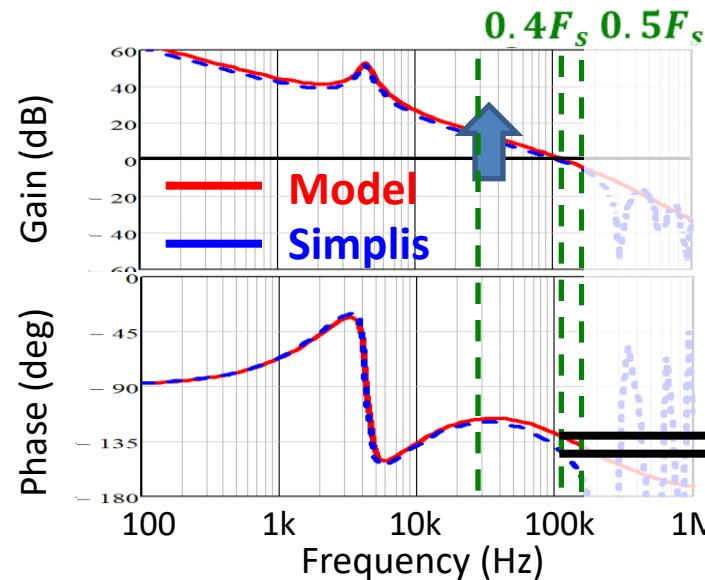
Effect of Bandwidth (High BW)



Closed-loop gain $\frac{v_B}{v_A}$

(Bandwidth = $0.4F_s$)

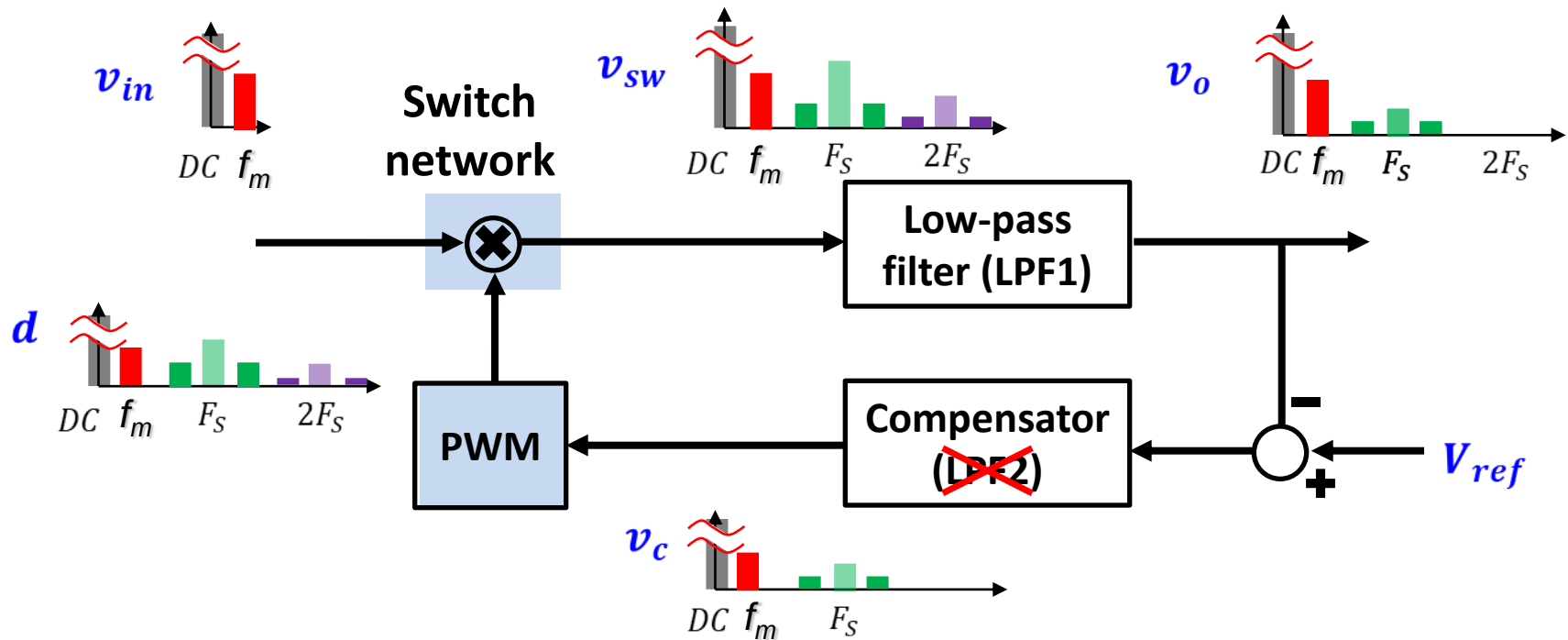
why increase bandwidth, the model is not accurate?
the error comes from powerstage or modulator



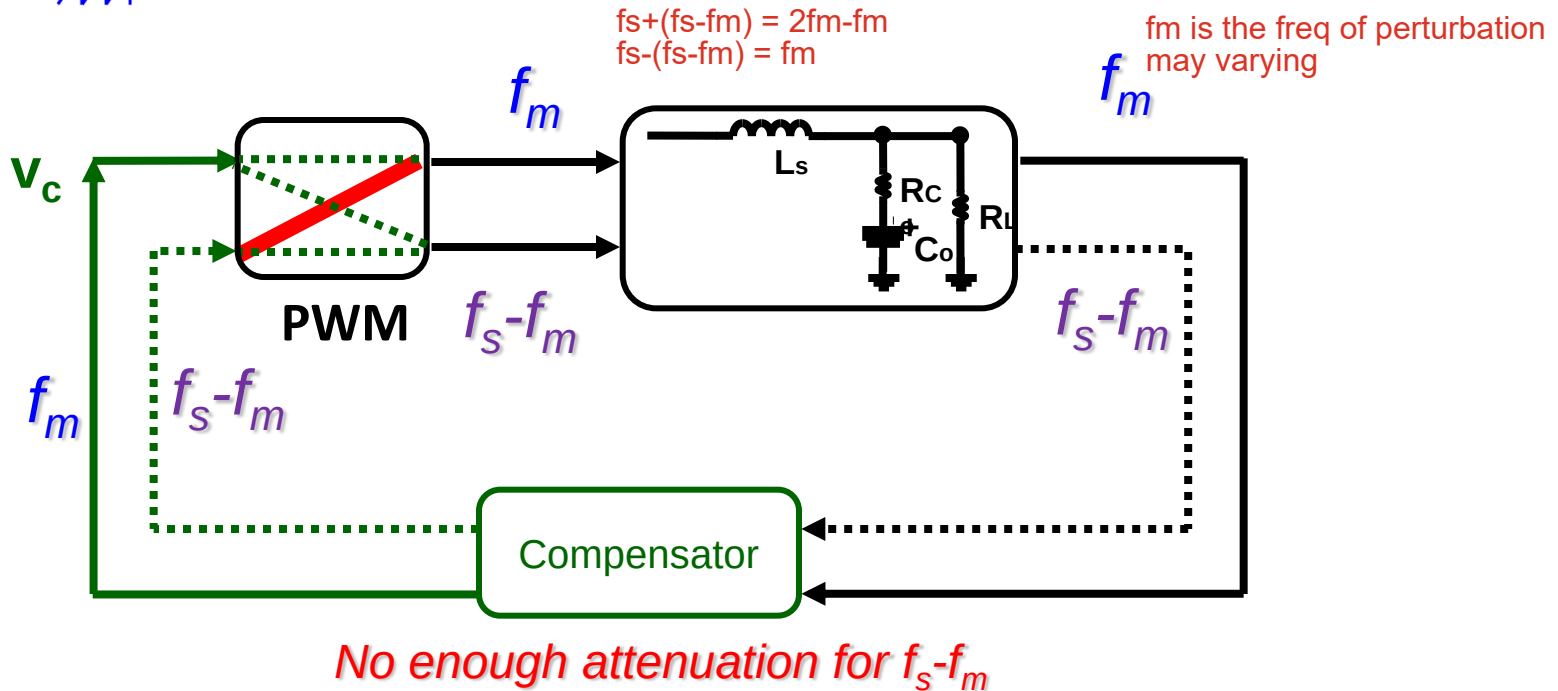
15° less phase margin than predicted

There will be some phase margin error with high bandwidth design

Limitation of the Average Model

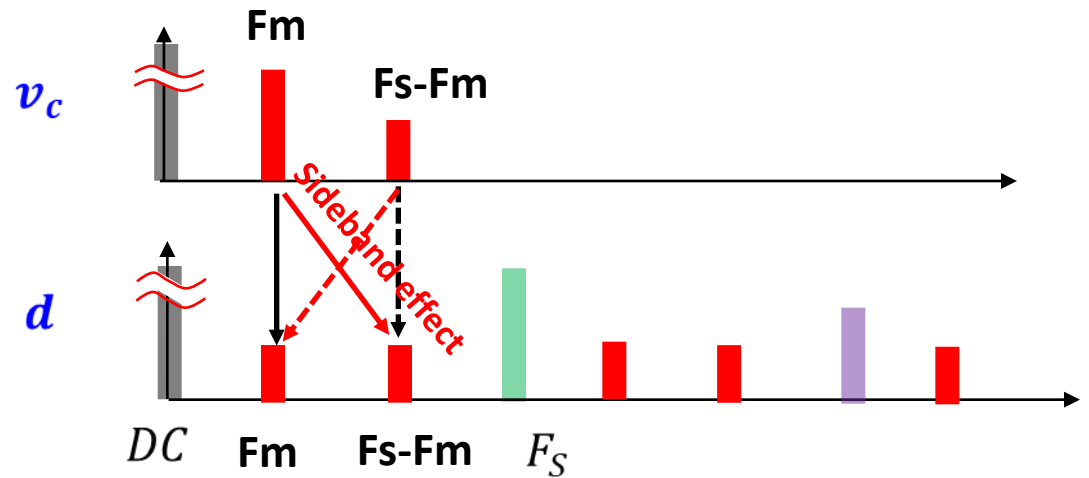
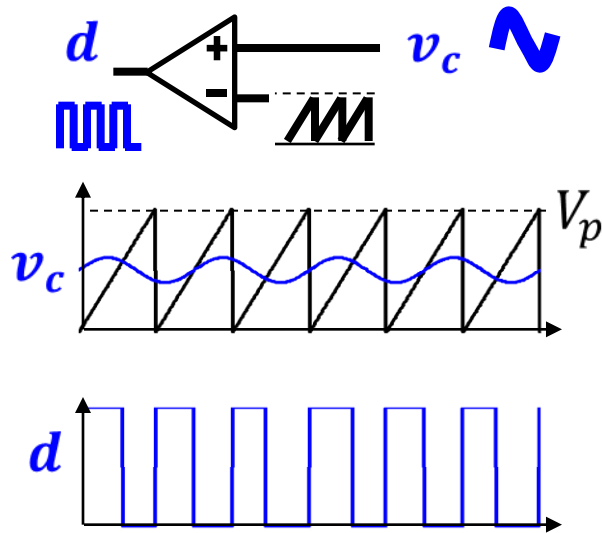


If compensator doesn't have strong low pass filter effect, v_c will also have some high frequency components



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Effect of $F_s - F_m$

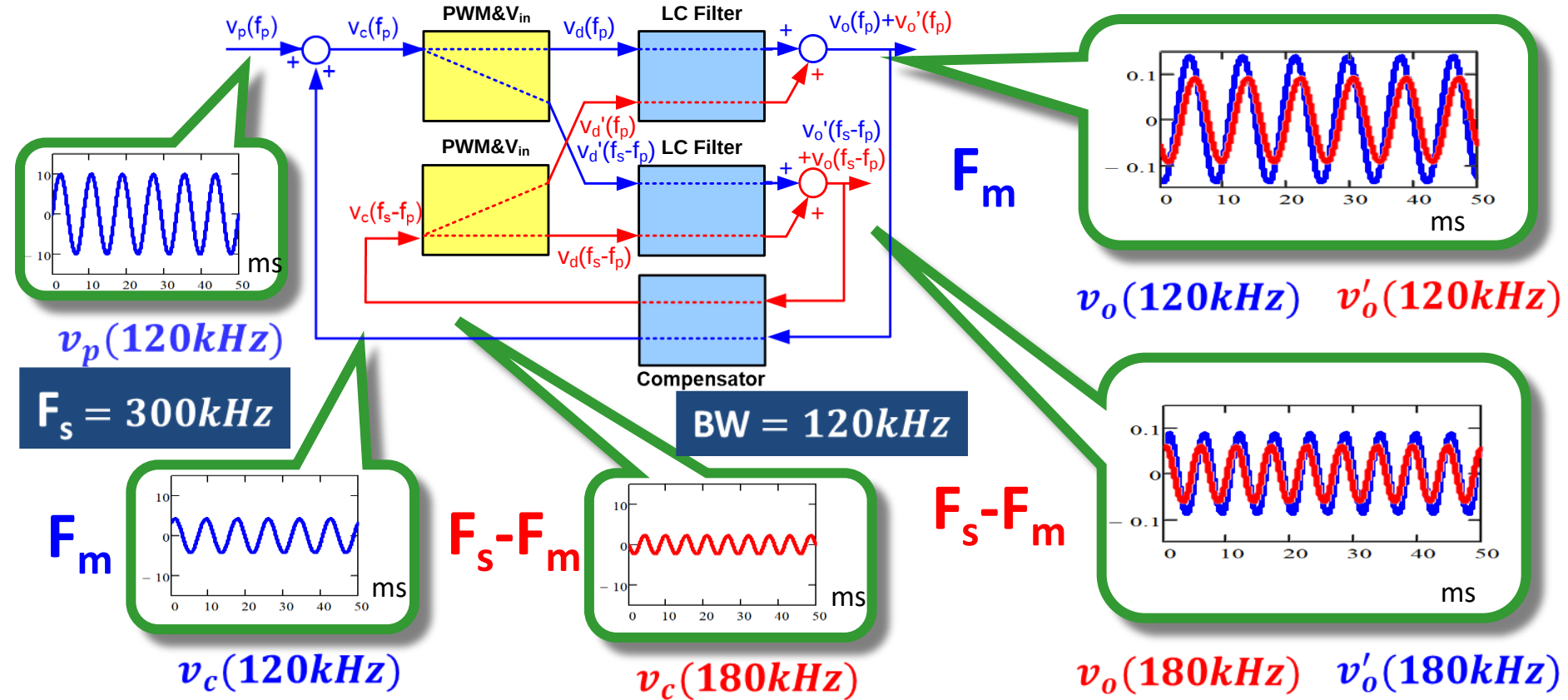


The $F_s - F_m$ component of v_c will also impact the F_m component of d



Scenario in High-Bandwidth Voltage-Mode Control

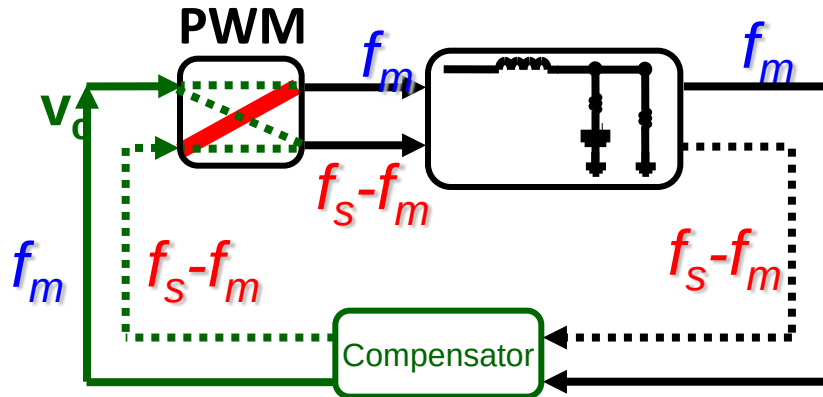
Yang Qiu (2005)



When BW is high, the contribution from $F_s - F_m$ cannot be ignored.

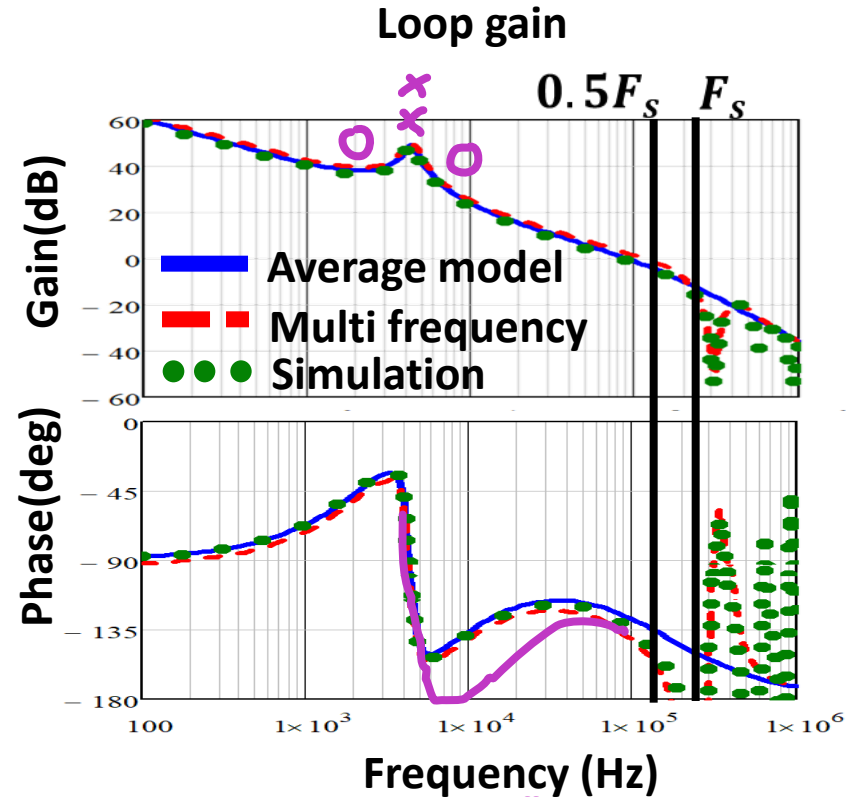
Multi-Frequency Modeling

Yang Qiu (2005)



Qiu's model
$$T_v(\omega_m) = \frac{T_{av}(F_m)}{1 + T_{av}(F_m - F_s)}$$

where $T_{av}(F_m)$ is loop gain from average model



The phase margin is not sufficient and there may exist low freq oscillation. This is not desired for design

The discrepancy mainly results from the first sideband $F_s - F_m$